Reading Notes: Food-101 and FoodSG-233

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What the two food image datasets contain, where they fall short, and how MealSenseAI combines them.

Why these datasets

These notes were written on 15 June 2026 while scoping Food Recognition for MealSenseAI. A photo-based meal logger is only as good as the dishes its classifier has seen, so the choice of training data comes first.

Food-101

Food-101 was published by Bossard, Guillaumin and Van Gool at the European Conference on Computer Vision. It contains one hundred and one food categories with a thousand images each: seven hundred and fifty for training and two hundred and fifty for testing. The training images were deliberately left uncleaned, so some label noise is expected and Model Evaluation should rely on the cleaner test split.

Its coverage is international: pizza, sushi, steak, ramen. Singapore hawker dishes are almost absent, which matters for a local audience.

FoodSG-233

FoodSG-233 covers two hundred and thirty-three dishes commonly eaten in Singapore, including chicken rice, laksa, nasi lemak and kaya toast. It fills exactly the gap that Food-101 leaves.

Combining both datasets gives a merged taxonomy that is broad and local at the same time. The cost is class imbalance, so Model Evaluation must report macro-F1 alongside top-1 accuracy, not accuracy alone.

Illustrative sample document for the CEREBRO demo library.